Although two facilities are sufficient to detect a gravitational wave, at least three and preferably four are required, at widely separated sites, to fully decode the wave's symphony, that is, to extract all the information the wave carries. A joint French/Italian team will build the third facility, named VIRGO,[11] near Pisa, Italy. VIRGO and LIGO together will form an international network for extracting the full information. Teams in Britain, Germany, Japan, and Australia are seeking funds to build additional facilities for the network.

It might seem audacious to construct such an ambitious network for a type of wave that nobody has ever seen. Actually, it is not audacious at all. Gravitational waves have already been proved to exist by astronomical observations for which Joseph Taylor and Russell Hulse of Princeton University won the 1993 Nobel Prize. Taylor and Hulse, using a radio telescope, found two neutron stars, one of them a pulsar, which orbit each other once each 8 hours; and by exquisitely accurate radio measurements, they verified that the stars are spiraling together at precisely the rate (2.7 parts in a billion per year) that Einstein's laws predict they should, due to being continually kicked by gravitational waves that they emit into the Universe. Nothing else, only tiny gravitational-wave kicks, can explain the stars' inspiral.

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What will gravitational-wave astronomy be like in the early 2000s? The following scenario is plausible:

By 2007, eight interferometers, each several kilometers long, are in full-time operation, scanning the skies for incoming bursts of gravitational waves. Two are operating in the vacuum facility in Pisa, Italy, two in Livingston, Louisiana, in the southeastern United States, two in Hanford, Washington, in the northwestern United States, and two in Japan. Of the two interferometers at each site, one is a "workhorse" instrument that monitors a wave's oscillations between about 10 cycles per second and 1000; the other, only recently developed and installed, is an advanced, "specialty" interferometer that zeroes in on oscillations between 1000 and 3000 cycles per second.